

Evaluating the Impact of Medicaid Expansion on Maternal Mortality & Perinatal Outcomes

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Abstract

We study whether Medicaid expansion under the Affordable Care Act affected maternal mortality rates in the United States. Using CDC WONDER mortality data from 2010–2020 and Medicaid expansion adoption dates from the Kaiser Family Foundation, we construct a state-year panel and estimate staggered difference-in-differences effects following Callaway and Sant’Anna (2021), with additional specifications that incorporate race-disaggregated mortality data. Across expansion cohorts, we find small and statistically insignificant average treatment effects, and event-study estimates show no consistent post-expansion decline in maternal mortality. Border-state difference-in-differences comparisons yield similarly imprecise estimates, highlighting the empirical difficulty of detecting policy effects in a rare and volatile outcome.

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1 Introduction

The decrease in the maternal mortality rate in the last century among high-income countries represents a drastic improvement of modern medicine and obstetric care. Across developed countries, the maternal mortality rate dropped from approximately 300-1000 deaths per 100,000 live births to between 1-20 per 100,000 live births (Neggers, 2016). However, the United States consistently ranks the highest in maternal mortality among developed countries (Center for Disease Control and Prevention, 2019; Kassebaum et al., 2013). Between 1990 and 2013, the maternal mortality rate in the U.S increased by 1.7% each year, in contrast to a global annual decline of 1.3% and a 3.1% decline among other developed countries (Kassebaum et al., 2013). Thus, despite significant improvement in medical technology and maternal care quality, the United States remains as one of the few countries worldwide in which the maternal mortality rate has steadily increased in the past 25 years (Coeytaux, Bingham, and Langer., 2010; Alkema et al., 2016; Center for Disease Control and Prevention, 2013).

Understanding maternal mortality requires attention to the health conditions that contribute to pregnancy-related deaths during both pre-conception and postpartum periods. Based on the data collected by U.S Department of Health and Human Services, Center for Disease Control Prevention (CDC), the leading causes of maternal mortality between 2006 and 2007 were cardiovascular conditions (13.5%), infections and hypersensitive disorders of pregnancy (11.1%), and anesthesia complications (6%) (Reid, 2013). Other factors, such as postpartum hemorrhages, preeclampsia, eclampsia and the increased operation of cesarean sections, are associated with higher maternal mortality rate (Reid, 2013).

However, the impact of these biological risk factors is not uniform across all women in the United States. While maternal mortality rates disproportionately affect lower income women, women of color experience higher maternal mortality rate across all income levels (Markides, 2011). Specifically, the maternal mortality rate is consistently the highest among African American women as they are twice more likely to die from pregnancy-related issues than the national average mortality rate for all women (Clark, Barger, and Corcoran, 2021). These authors point out that, beyond preexisting health conditions (e.g., hypertension, diabetes, and obesity) and postpartum complications, African American women are subjected to systematic racial disparities in healthcare access that drastically increase their risks of maternal mortality. Notably, as they are more likely to reside within areas with limited healthcare resources and infrastructures, African American women experience prevalent delay in prenatal care during the first trimester and receive lower-quality overall healthcare during their pregnancies.

Enacted on March 23, 2010, the Affordable Care Act (ACA), signed by President Obama into law, marks a major U.S federal health reform in expanding healthcare coverage, improving cost affordability, and enhancing care quality. Prior to the ACA, since 1989, federal laws have required states to provide Medicaid coverage to any pregnant citizen whose family income was below or at 133% of the federal poverty level for the duration from conception up to 60 days postpartum (Medicaid and CHIP payment and Access Commission, 2016). However, despite the national mandate, Medicaid coverage for nondisabled, nonpregnant low-income adults was restricted and varied across states. Notably, between 2005 and 2013, approximately 55% of individuals covered by Medicaid at delivery experienced a lapse in coverage within six months after delivery (Bellerose, Collin, and Daw, 2022). The instability of healthcare coverage has contributed significantly to worse maternal and child outcomes. Specifically, women’s loss of healthcare access after 60 days can lead to increased risks of postpartum health complications (Eliason, 2019). Other factors such as substance overuse, postpartum depression, and untreated perinatal mood disorders, are common drivers of maternal mortality within the first year after birth (Clark, Barger, and Corcoran, 2021). Starting in 2014, under ACA, states were encouraged to implement Medicaid expansion for households whose income fell below 138% of the Federal Poverty level (Bellerose, Collin, and Daw, 2022). Thus, the policy for Medicaid expansion did not affect the original eligibility during pregnancy but extended coverage into the preconception period and beyond 60 days postpartum (Bellerose, Collin, Daw, 2022). Based on the variations of the states’ adaptations under ACA, this research project aims to examine the extent to which the Medicaid expansion under the 2010 Affordable Care Act (ACA) impacts maternal mortality rates in the United States while controlling for relevant factors.

2 Literature Review

A growing body of research has examined whether Medicaid expansion under the ACA influences maternal health outcomes. Eliason (2020) used a difference in differences approach to compare maternal mortality ratios in states that adopted Medicaid expansion with those that did not during the study period (2006-2017). Using Centers for Disease Control and Prevention WONDER Online Database, the authors define maternal mortality in two ways. The first definition captures maternal deaths occurring during pregnancy, within 42 days of delivery and during the late postpartum period. The second definition excluded late maternal deaths, allowing the study to assess whether the reduction in maternal mortality following Medicaid expansion are driven by improved postpartum coverage or by changes in preconception insurance access. Results indicate that Medicaid expansion is associated

with approximately 7 fewer maternal deaths per 100,000 live births, with similar reductions when late maternal deaths are excluded. The authors suggest that expanding insurance coverage before and after pregnancy may contribute to improved maternal outcomes. Stratified analyses show that these reductions are concentrated among non Hispanic Black women, highlighting Medicaid’s potential role in reducing racial disparities in maternal mortality. However, the study exploits staggered state-level adoption of ACA Medicaid expansion by using a traditional two way fixed effects difference-in-difference model. This may bias estimates in the presence of heterogeneous treatment effects.

A more recent study examined the effect of state-level Medicaid expansion decisions on maternal health outcomes using 2018-2020 CDC WONDER and Healthcare Cost and Utilization Project (HCUP) data (Williams, A. M et al 2024). The authors compared maternal mortality and severe maternal morbidity across expansion and nonexpansion states while controlling for race, ethnicity, and geographic region. They applied the same ICD-10 codes as Eliason et al. when defining maternal death though their central method of analysis was a two-tailed non-paired-t-test. Main findings highlight that maternal mortality rates were approximately 5 fewer deaths per 100,000 live births in Medicaid expansion states compared to nonexpansion states (20.7 vs 26.0) while SMM rates were higher in expansion states. Parallel to Eliason’s finding, maternal mortality remained substantially higher among Black and Hispanic women than among White women in both expansion and nonexpansion states. In their analysis of geographical location, the authors observed that urban states that expanded Medicaid experienced lower maternal mortality than urban nonexpansion states (22.6 vs 27.0 deaths per 100,000 live births). Key limitation to this study involves data suppression and inconsistent methods used across states to report maternal mortality and morbidity. Additionally, SMM morbidity data were restricted to in-hospital events though a large percentage of SMM occurs after discharge. Finally, the use of a two-tailed, non-paired t-test assumes independence between states and normal distribution of maternal mortality rates, which may not fully account for state-level clustering, population differences or potential confounding variables, limiting the robustness of the statistical comparison.

A related study used a difference in differences design with data from the American Community Survey to examine how Medicaid expansion affected insurance coverage among low income adult women (Bullinger et al 2022). Focusing on ages 18-44, the authors compared women without children, postpartum mothers, and mothers whose youngest child was older than one year across expansion and nonexpansion states. Based on their analyses, the expansion increased coverage among adult women with incomes between 101 and 200% of the federal poverty line (FPL) without children by 10.7 percentage points. Coverage of mothers with children older than one year increased by 9.5 percentage points and coverage of

mothers with infants rose by 7.9 percentage points. As such, results imply that Medicaid expansion produced the largest coverage gains for low income women without children, whereas postpartum mothers saw the smallest increases in coverage. Mothers of older children experienced moderate gains. While the difference-in-difference approach strengthens the causal inference, the analysis relies on self-reported insurance coverage from the ACS, which may be subject to reporting error. An additional shortcoming of this study is the relatively short post-expansion period examined as coverage gains appear to plateau or decline toward the end of the study window.

Our research builds upon this prior research and makes several important contributions. First, like Eliason (2020), we use CDC WONDER mortality data and the same ICD-10 codes to construct state-year maternal mortality rates. We examine the differences between Medicaid expansion and nonexpansion states at the state-year level. Second, we extend Eliason (2020) and Williams et al (2024) by employing a more modern staggered difference-in-differences framework following Callaway and Sant’Anna Staggered, which explicitly accounts for variation in the implementation of Medicaid expansion across states. Finally, findings from Bullinger et al 2022 help contextualize the pathway through which Medicaid expansion may influence maternal outcomes. Given that insurance coverage gains differed between maternal subgroups, we interpret the estimated effects of Medicaid expansion on maternal mortality with caution.

3 Data Sources

3.1 Mortality Data

Our primary outcome of interest is the maternal mortality rate, which is commonly measured as maternal deaths per 100,000 population. Mortality counts are obtained from the CDC WONDER Cause of Death database, which provides annual counts of maternal death and population denominators by geographic unit. Mortality measures will be based on ICD-10 codes, specifically O00-O99 (maternal causes of death). We extract state-level maternal mortality data for the period 2010-2020, corresponding to the years surrounding major Medicaid expansion decisions under the ACA. Because CDC WONDER mortality files provide population denominators but not live birth counts, we compute maternal mortality as deaths per 100,000 population. To construct a consistent outcome variable across states and years, we compute the maternal mortality rate manually as:

$$\text{MMR}_{st} = \frac{\text{Maternal Deaths}_{st}}{\text{Population}_{st}} \times 100,000 \quad (1)$$

where s indexes state and t indexes years. This approach avoids reliance on CDC WONDER’s reported crude rates, which are sometimes flagged as unreliable due to small cell counts. All observations with missing deaths or population counts are dropped prior to analysis. The resulting dataset forms an unbalanced state-year panel with 48 units, as Wyoming and Vermont are consistently suppressed in CDC WONDER, and Alaska reports insufficient reliable observations for inclusion across the full period. Furthermore, we later include race in this dataset, which CDC WONDER groups as a state-year-race panel observed over 11 years.

3.2 Medicaid Expansion Data

Information on Medicaid expansion timing is drawn from publicly available data compiled by the Kaiser Family Foundation (KFF). This dataset provides information on whether the state adopted Medicaid expansion and the exact date of implementation. Using the KFF data, we construct two key variables: ‘Adopted’, an indicator equal to one if a state ever adopted Medicaid expansion during the sample period, and zero otherwise. ‘Expansion year’, defined as the first year in which Medicaid expansion took effect in that state. States that never expanded Medicaid during the sample window are coded as never-treated, with missing expansion years. These states serve as part of the comparison group in the staggered difference-in-difference framework.

3.3 Merged Analysis Sample

The CDC WONDER data are merged with the KFF Medicaid expansion data using standardized state names. Observations that can not be matched across datasets are dropped. The final analysis consists of a state-by-year panel containing maternal mortality outcomes, population counts, and Medicaid expansion timing. We include a separate panel containing state-by-year-by-race data which compresses the number of total observations, as the race of individuals who suffered maternal death is not always reported in CDC data. These two merged datasets allow for descriptive comparisons between the expansion and non-expansion states.

3.4 Descriptive Statistics and Trends

The first table presents summary statistics for the state-year panel. Across 559 state-year observations, the average maternal mortality rate is approximately 3.9 deaths per 100,000, with substantial variation across states and time. Death counts also exhibit wide dispersion.

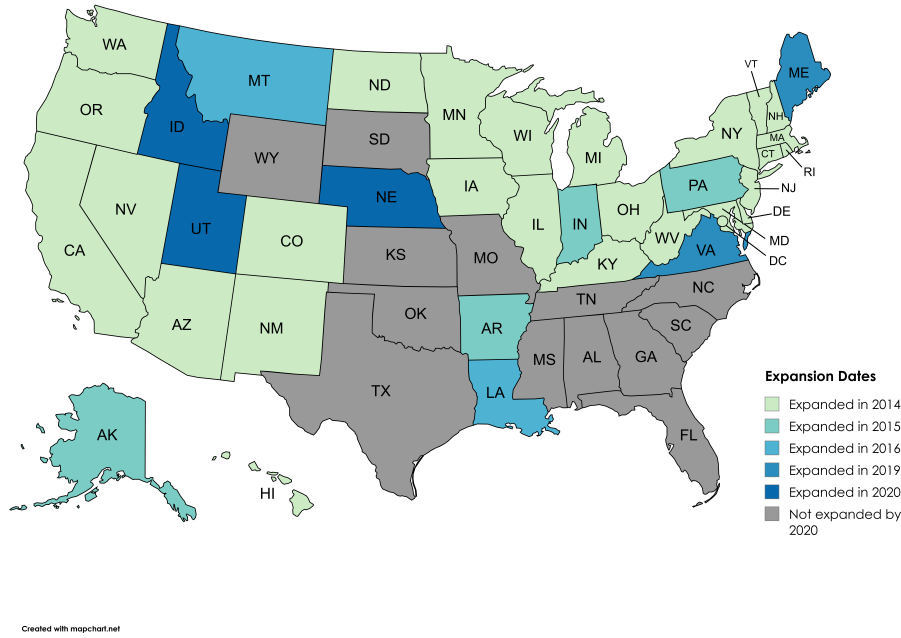


Figure 1: Expansion Status by Expansion Year

Table 2 summarizes the state-year-race panel, reporting counts, population shares, and the MMR distributions by race category. Black or African American women account for 46% of the state-year observations in the race panel and experience substantially higher average maternal mortality rates than other groups.

Table 1: Summary Statistics: State-Year Panel

	Observations	Mean	SD	Min	Max
Maternal mortality rate (per 100k)	559	3.888512	1.035697	1.597416	9.223361
Maternal deaths (count)	559	245.2844	263.4917	10	1346
Population (denominator)	559	6,302,588	7,099,133	563,626	3.96e+07
N	559				

Notes: Unit of observation is state-year (2010–2020). MMR is maternal deaths per 100,000.

Figure 2 plots average maternal mortality rates over time for Medicaid expansion versus non-expansion states. While these trends are not causal, they provide useful context. Expansion states exhibit a gradual decline in maternal mortality over the sample period, whereas non-expansion states display higher average rates and greater volatility, particularly in later years.

Table 2: Race Composition and Summary Statistics: State-Year-Race Panel

Race	Count	Share (%)	Mean MMR	SD MMR	Min MMR	Max MMR
Asian or Pacific Islander	23	4.3	3.284	1.678	1.139	7.692
Black or African American	249	46.1	10.735	3.536	3.927	30.744
White	268	49.6	3.938	1.460	1.570	11.517
<i>N</i>	540	100.0				

Notes: Count is the number of state-year observations within each race category. MMR denotes maternal deaths per 100,000.

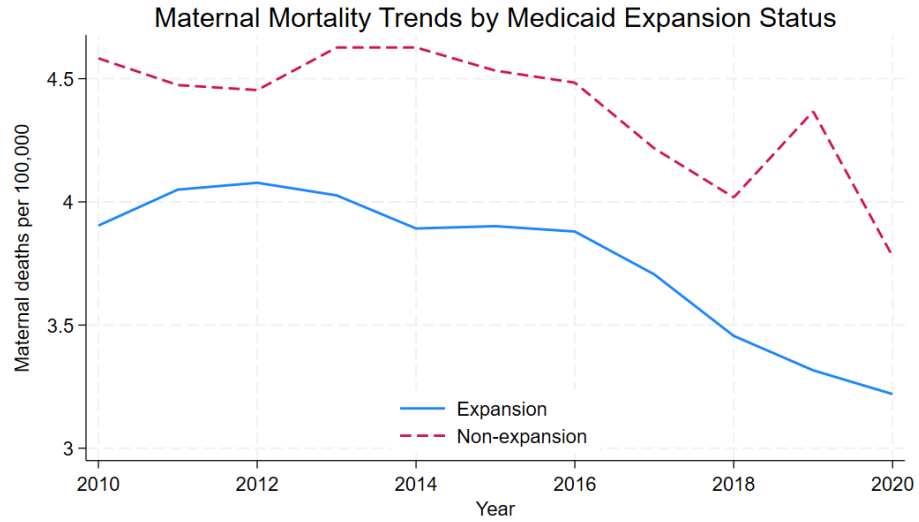


Figure 2: Maternal Mortality Trends by Expansion Status Over Study Period

3.5 Data Limitations

Several limitations of the data require careful acknowledgement. First, maternal mortality is a very rare event, particularly at finer geographic or demographic levels, which introduces substantial year-to-year volatility and limits statistical power. Second, the CDC WONDER data are subject to suppression and reliability concerns for small cells, which constrains our ability to consistently observe county-level outcomes and motivates our primary focus on state-year panels. Third, while our race-specific data provide important descriptive context, racial classifications are aggregated and do not capture heterogeneity within groups or differences in reporting quality across states and over time. Finally, while we explored the possibility of incorporating measures of healthcare provider quality or hospital capacity as additional controls, we were unable to identify a publicly available, consistently measured dataset at the state or county level over this period. As a result, the analysis focuses on demographic and policy variation, and unobserved differences in healthcare quality may remain a source of residual confounding.

4 Empirical Methods

4.1 Empirical Setup and Units of Analysis

Our primary analysis is conducted using a state-year panel spanning 2010-2020, where the unit observation is a U.S. state in a given year. The outcome variable is the maternal mortality rate (MMR). Treatment timing is defined using the first year in which a state implemented Medicaid expansion. States that never expanded Medicaid in the study period are treated as never-treated units.

However, to address concerns about compositional differences across states and over time, we construct a state-year-race panel, where observations are disaggregated by major racial groups (White, Black or African American, Asian or Pacific Islander). We will include race fixed effects to control for persistent differences in maternal mortality across racial groups. The race-disaggregated panel therefore will serve as a validity check on the aggregate state-level estimates.

4.2 Staggered Difference-in-Differences Framework

Because Medicaid expansion was adopted at different times across states, we rely on a staggered difference-in-differences method that is robust to heterogeneous treatment timing and treatment effects. Our main estimator follows Callaway and Sant’Anna (2021) which

estimates group-time average treatment effects on the treated (ATTs) by comparing treated states to appropriate control groups consisting of never treated and not-yet-treated states.

Let Y_{it} denote the maternal mortality rate in state i at time t , and let G_i denote the year in which state i first adopts Medicaid expansion. Then the group-time ATT is defined as the following:

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g] \quad (2)$$

where $Y_t(g)$ is the potential outcome if a state is treated starting in year g , and $Y_t(0)$ is the potential outcome if the state is never treated. This allows treatment effects to vary flexibly across cohorts and over time, avoiding bias that arises in conventional two-way fixed effects models when treatment effects are heterogeneous. We implement this estimator using the ‘csdid’ command in Stata. Standard errors are clustered at the state level, and event-study estimates derived from the group-time ATTs are used to assess pre-treatment trends.

4.3 Border-State Difference-in-Differences Analyses

Due to the noise present in our state-year panel data, we also conduct border-state DiD comparisons using selected pairs of geographically adjacent states with similar regional and demographic characteristics but different Medicaid expansion timing. The border pairs meeting this criterion were Oklahoma-Arkansas, Minnesota-Wisconsin, and Kentucky-Tennessee.

These analyses restrict the sample to a small number of neighboring states and estimate a standard DiD model comparing outcomes before and after expansion in the treated state relative to its bordering control state. While these comparisons rely on smaller samples than the DiD framework, they control for regional factors that evolve across state borders. Adjacent states are more likely to share similar labor markets, health care referral networks, population health trends, and exposure to regional economic shocks, reducing concerns that treatment effects are driven by broad geographic differences. For each border pair, we estimate a standard DiD model of the form:

$$Y_{st} = \alpha + \beta(\text{Treat}_s \times \text{Post}_t) + \delta_s + \lambda_t + \varepsilon_{st} \quad (3)$$

where Y_{st} denotes the maternal mortality rate in State s and year t , Treat_s indicates the treated state within the pair, and Post_t equals one in years following Medicaid expansion in the treated state. State and year fixed effects control for time-invariant differences between states and common national shocks, respectively. The coefficient of interest, β , represents the change in maternal mortality in the expansion state relative to the neighboring control state after expansion.

5 Results

5.1 Unadjusted Staggered Difference-in-Differences Analyses

We begin by presenting unadjusted staggered Difference-in-Differences estimates of the effect of Medicaid expansion on maternal mortality rates using the state-year panel. All coefficients are interpreted as absolute changes in maternal deaths per 100,000 population. Table 3 reports pre- and post-treatment averages constructed from the event-study coefficients produced by `estat event`. Event-study estimates by year of Medicaid expansion provide little evidence of systematic heterogeneity in post-treatment dynamics. At the aggregate level, the parallel trends assumption is satisfied in our analysis. The average pre-treatment difference between the expansion and non-expansion states is statistically indistinguishable from 0 (Table 4: `pre_avg` = -0.002 , $p = 0.978$), indicating that the treated and control states followed similar trends prior to expansion.

In all expansion cohorts, the overall estimated average effect of treatment on the treated (ATT) is small in magnitude and statistically insignificant, with an average increase of 0.089 maternal deaths per 100,000 in the post-expansion period, relative to the counterfactual (Table 4). This suggests that there is no detectable average change in maternal mortality rates after Medicaid expansion when aggregating between states and years.

Table 3: Staggered DiD Event Study: Pre- and Post-Treatment Averages

	ATT	SE	p-value	95% CI (LB)	95% CI (UB)
Pre-treatment average	-0.002	0.058	0.978	-0.116	0.113
Post-treatment average	0.007	0.131	0.955	-0.249	0.264

Notes: Pre- and post-treatment averages are computed from the event-study coefficients reported by `estat event`.

Table 4: Staggered DiD (`csdid`): Overall ATT

	ATT	SE	p-value	95% CI (LB)	95% CI (UB)
Overall ATT	0.089	0.140	0.522	-0.184	0.363

Notes: Overall ATT from the Callaway–Sant’Anna staggered DiD estimator. Unit of observation is state $\times$ year. Outcome is maternal deaths per 100,000.

To assess dynamic treatment effects, Table 5 and Figure 3 present event-study estimates of the ATT by event time and by calendar year. Pre-expansion coefficients are generally close to zero and statistically insignificant, providing some reassurance that treated and comparison states followed similar trends prior to expansion. However, post-expansion estimates also

Table 5: Event Study (csdid): Dynamic ATT by Event Time

Event time	ATT	SE	p-value	95% CI (LB)	95% CI (UB)
0	0.098	0.153	0.524	-0.203	0.398
1	-0.044	0.173	0.136	-0.382	0.294
2	0.082	0.222	0.528	-0.353	0.517
3	-0.062	0.187	0.753	-0.428	0.304
4	0.002	0.303	0.770	-0.593	0.597
5	0.133	0.271	0.651	-0.398	0.664
6	-0.109	0.357	0.712	-0.810	0.592
7	0.552	0.499	0.269	-0.426	1.530
8	-0.143	0.336	0.670	-0.802	0.516
9	0.001	0.413	0.998	-0.809	0.811
10	-0.554	0.235	0.018	-1.013	-0.094

Notes: Event time is years relative to each state's Medicaid expansion year (0 = expansion year). When multiple estimates exist for the same event time, we report the mean across those estimates for readability.

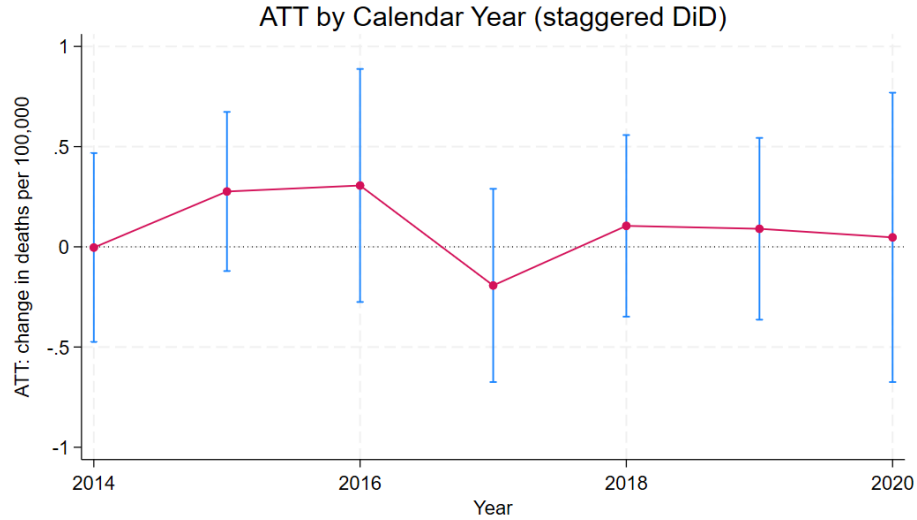


Figure 3: Average Treatment Effects by Calendar Year (staggered DiD)

exhibit substantial variability and wide confidence intervals, with no consistent downward pattern in maternal mortality.

5.2 Staggered Difference-in-Differences Estimates with Race Controls

Given substantial racial disparities in maternal mortality, we extend the staggered DiD analysis to incorporate race as an explicit control using a state $\times$ race $\times$ year panel. Table 6 reports the overall race-adjusted ATT. The estimated effect remains statistically insignificant, with an estimate of 0.060 less maternal deaths per 100,000 in the post-expansion period, and a wide confidence interval. Relative to the unadjusted specification, the estimate is slightly more negative, suggesting that accounting for racial heterogeneity may reduce some noise present in aggregate state-level estimates.

Table 6: Staggered DiD (csdid): Overall ATT with Race Controls

	ATT	SE	p-value	95% CI (LB)	95% CI (UB)
Overall (race-adjusted)	-0.060	0.346	0.862	-0.738	0.617

Notes: Unit of observation is state $\times$ race $\times$ year. Race is controlled for via race fixed effects (included in the outcome model) within the staggered DiD setup.

Table 7: Staggered DiD (csdid): Race-Specific ATT

	ATT	SE	p-value	95% CI (LB)	95% CI (UB)
Black or African American	0.029	0.566	0.960	-1.081	1.138
White	-0.086	0.332	0.796	-0.738	0.566

Notes: Each row reports an overall ATT from a separate `csdid` estimation within that race group. Unit of observation is state $\times$ year within race.

Event-study plots by race (Figure 4) show no clear pre-trend violations but also no consistent post-expansion reductions in maternal mortality. Together, these results highlight both the large baseline racial disparities in maternal mortality and the empirical difficulty of detecting modest policy effects in a rare outcome using state-level panels.

5.3 Border-State Difference-in-Differences Analyses

We next estimate a standard two-state DiD model comparing MMR before and after expansion in the treated state relative to its neighboring control state. The resulting estimates (Table 8) vary in sign and magnitude across pairs and are generally imprecise.

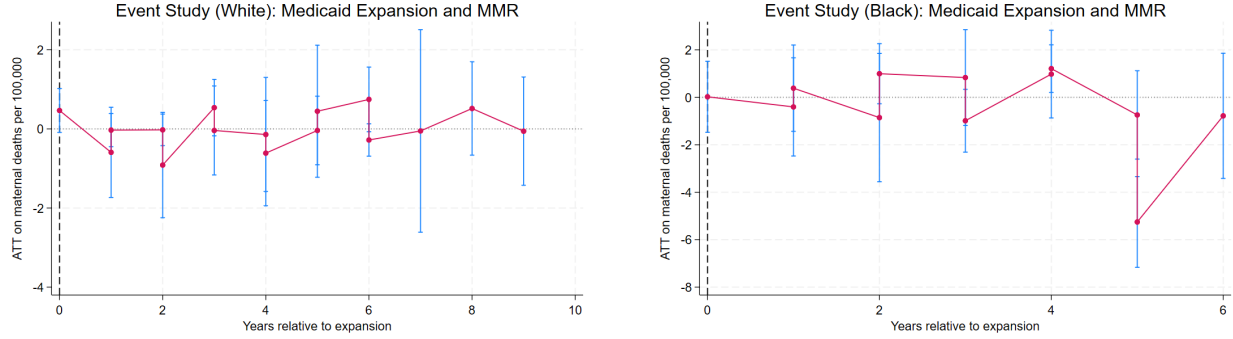


Figure 4: Race-Specific Event Study Estimates

Table 8: Border-State Difference-in-Differences Estimates

Border pair	DiD estimate	SE	p-value	95% CI (LB)	95% CI (UB)
Arkansas (treated) vs Oklahoma	-0.105	0.565	0.857	-1.213	1.003
Minnesota (treated) vs Wisconsin	0.296	0.259	0.283	-0.212	0.804
Kentucky (treated) vs Tennessee	-0.363	0.287	0.237	-0.926	0.199

Notes: Each row reports the coefficient on a treated $\times$ post indicator from a separate two-state DiD with state and year fixed effects. Standard errors are heteroskedasticity-robust (not clustered) because each comparison includes only two states.

In the Arkansas–Oklahoma comparison, Arkansas is treated (expanded in 2014) and Oklahoma expanded later; the estimate implies a -0.105 change in maternal deaths per 100,000 in Arkansas relative to Oklahoma after 2014. For Minnesota–Wisconsin, Minnesota is treated (expanded in 2014) while Wisconsin did not expand in the study period; the estimate implies a 0.296 increase in maternal deaths per 100,000 in Minnesota relative to Wisconsin after 2014. For Kentucky–Tennessee, Kentucky is treated (expanded in 2014) while Tennessee did not expand in the study period; the estimate implies a -0.363 change in maternal deaths per 100,000 in Kentucky relative to Tennessee after 2014.

6 Discussion

The empirical analyses presented above yield largely inconclusive estimates of the effect of Medicaid expansion on maternal mortality, both in the staggered difference-in-differences framework and in the border-pair design. While point estimates generally align with the hypothesis that Medicaid expansion may reduce maternal mortality, they are imprecisely estimated and statistically indistinguishable from zero across most specifications. Rather than rejecting the relevance of Medicaid expansion for maternal health outcomes, these findings highlight the substantial econometric and data challenges inherent in estimating

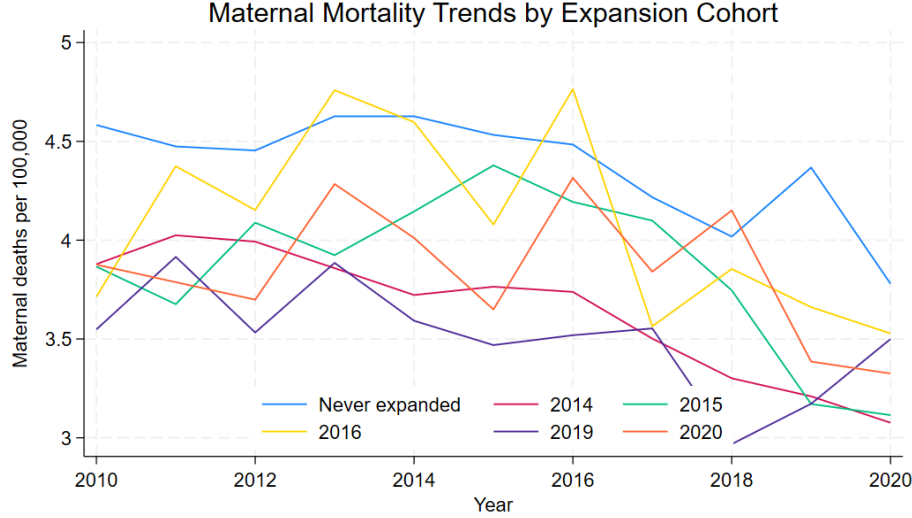


Figure 5: Maternal Mortality Trends by Expansion Cohort

causal effects on rare, high-variance outcomes such as maternal mortality.

A central identifying assumption of the staggered difference-in-differences design is that, absent treatment, treated and untreated units would have followed parallel trends. When assessed in aggregate, standard pre-treatment event-study diagnostics suggest that this assumption is reasonably satisfied. However, this aggregate validation may mask meaningful heterogeneity across expansion cohorts. Medicaid expansion occurred at different times and under different political, institutional, and healthcare market conditions across states. When the parallel trends assumption is disaggregated by adoption year the plausibility of a common counterfactual trajectory becomes less clear as seen in Figure 5. Differential pre-trends across expansion cohorts can bias pooled DiD estimates, even when overall diagnostics appear well behaved. In this sense, the staggered nature of the policy itself may cloud inference, as cohort-specific deviations from parallel trends are averaged into a single estimate, potentially attenuating true effects.

The border-pair analysis was intended to strengthen identification by comparing geographically proximate states exposed to similar socioeconomic environments but different Medicaid expansion regimes. While this design improves comparability along observable dimensions, it does not fully resolve concerns about unobserved heterogeneity. In particular, hospital quality represents a key omitted variable that likely biases the estimated treatment effect. Hospital quality varies systematically across space and is strongly correlated with maternal mortality outcomes, yet it is only imperfectly captured by publicly available controls such as urban status. Expansion states may differ from non-expansion states in baseline hospital capacity, obstetric care quality, or the presence of high-risk referral cen-

ters. If higher-quality hospitals are more prevalent in expansion states, failure to account for this heterogeneity may lead to upward bias in estimated effects; conversely, if expansion disproportionately occurred in states with weaker healthcare infrastructure, estimates may be biased downward. In either case, omitted hospital quality introduces ambiguity into the direction and magnitude of the estimated effect.

A further challenge arises from the rarity of maternal mortality itself. Maternal deaths are low-incidence events, particularly at the county or even state-year level, resulting in substantial sampling variability. Small changes in the number of deaths can generate large swings in mortality rates, amplifying noise relative to signal. This high variance reduces statistical power and increases the likelihood of Type II error, especially when treatment effects are modest or operate through indirect channels such as improved prenatal care or reduced financial stress. The combination of staggered treatment timing and low-incidence outcomes further complicates inference, as some expansion cohorts contribute relatively few post-treatment observations with non-zero events. As a result, estimates may be driven by a small number of observations, rendering them highly sensitive to specification choices. This is why our initial plan to use county level data for border pair analysis failed, it would have allowed us to remove OVB such as hospital quality but the variance was too random and too significant to get any meaningful analysis.

These challenges suggest several avenues for improving future analyses. First, greater attention to cohort-specific dynamics may yield more credible estimates. Second, incorporating richer measures of hospital quality, including private request data sources such as the Healthcare Cost and Utilization Project, would help address omitted variable bias. Where direct measures are unavailable, instrumental variable strategies or hospital fixed effects in more granular data may offer partial solutions. Third, aggregating outcomes over longer time windows or employing alternative maternal health measures, such as severe maternal morbidity, could mitigate the statistical limitations posed by low-incidence mortality data.

Taken together, the inconclusive results should not be interpreted as evidence that Medicaid expansion has no effect on maternal mortality. Rather, they reflect the difficulty of identifying causal effects in settings characterized by heterogeneous policy implementation, unobserved healthcare quality, and rare outcomes. By explicitly acknowledging these limitations, this study contributes to a more nuanced understanding of the empirical challenges in evaluating large-scale health policy reforms and points toward methodological refinements that may yield clearer insights in future work.

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